



JC4002 Knowledge Representation

Day 2: Expressing Knowledge

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Day 2

Expressing Knowledge

1. Knowledge Engineering
2. Vocabulary
3. Facts
4. Entailment

Knowledge Engineering

- We have learned what knowledge-based systems are and how they can be useful in AI
 - meaning and entailment
 - individuals and properties
 - encoding facts for entailment relations
- We have decided to use FOL as our initial representation language for knowledge
- Now that we know the basics, how can we design a KB from scratch?

Soap Opera Example

- ❑ Similar to any computer system, we must understand what is to be computed
- ❑ As well as why and where inference reasoning is necessary
- ❑ Example domain: *soap opera world*
 - Represent people, places, companies, marriages, divorces, deaths, kidnappings, crimes, etc.
- ❑ How can we design a KB with the appropriate entailment relations? What vocabulary do we use? Which facts do we represent?

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Vocabulary Overview

Create a domain-dependent vocabulary containing the non-logical symbols of our KB (predicates, functions, and constants)

- ☐ named individuals (constants)

- Alice, John, Aberdeen, UniversityOfAberdeen, . . .

- ☐ basic types (predicates)

- Person, Place, City, Woman, Man, . . .

- ☐ attributes (predicates)

- Rich, Smart, Beautiful, . . .

- ☐ relationships (predicates or functions)

- LivesAt, MarriedTo, DaughterOf, HadAffair, Blackmails, . . .

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Basic Facts

Database-like facts usually in the form of atomic sentences and negations

- type
 - Woman(Alice), Man(John), City (Aberdeen), Company (InsuranceCorporation)
- attribute
 - Rich(John),
 - $\neg$ HappilyMarried (Alice),
 - WorksFor(John, UoA)
- equality / constants
 - John = ceoOf(IC)
 - UoA = UniversityOfAberdeen
 - IC = InsuranceCorporation
 - bestFriendOf(Jim) = John

Complex Facts

Universal abbreviations, possible to express without quantifiers but very tedious and prone to error, as well as not future-proof

- $\forall x (\text{Woman}(x) \wedge \neg(x = \text{Alice}) \rightarrow \text{Loves}(x, \text{John}))$
- $\forall x (\text{Rich}(x) \wedge \text{Man}(x) \rightarrow \text{Loves}(x, \text{Alice}))$
- $\forall x \forall y (\text{Loves}(x, y) \rightarrow \neg \text{Blackmails}(x, y))$

Incomplete knowledge, not able to write down a more complete version

- $\exists x (\text{Adult}(x) \wedge \text{Blackmails}(x, \text{John}))$ who?
- $\text{Loves}(\text{Jane}, \text{John}) \vee \text{Loves}(\text{Jane}, \text{Jim})$ which?

Closure axioms are used to limit the domain of discourse

- $\forall x (\text{Person}(x) \rightarrow x = \text{Alice} \vee x = \text{John} \vee x = \text{Jim} \vee \dots)$
- $\forall x (x = \text{UoA} \vee x = \text{Alice} \vee x = \text{John} \vee x = \text{Jim} \vee \dots)$

Terminological Facts

Relationships among predicates may be required:

Disjoint	$\forall x \text{ Man}(x) \rightarrow \neg \text{Woman}(x)$
Subtype	$\forall x \text{ Senator}(x) \rightarrow \text{Legislator}(x)$
Exhaustive	$\forall x \text{ Adult}(x) \rightarrow \text{Man}(x) \vee \text{Woman}(x)$
Symmetry	$\forall x \forall y \text{ MarriedTo}(x, y) \rightarrow \text{MarriedTo}(y, x)$
Inverse	$\forall x \forall y \text{ ChildOf}(x, y) \rightarrow \text{ParentOf}(y, x)$

Activity 1

In pairs, discuss the following question (5 minutes) and then share your answers with the rest of your colleagues (5 minutes):

“When translating ‘every women loves john’, why does the translation not become $\forall x \text{ Woman}(x) \wedge \text{Loves}(x, \text{John})$?”

Symmetry Example

Recall example from previous lecture, now with a new sentence added:

1. $\text{Prof}(\text{Alice})$
2. $\forall x (\text{Prof}(x) \rightarrow \text{Person}(x))$
3. $\text{Dean}(\text{John})$
4. $\forall x (\text{Dean}(x) \rightarrow \text{Prof}(x))$
5. $\forall x \exists y (\text{Prof}(x) \wedge \text{Dean}(y) \rightarrow \text{Friend}(x, y) \vee \neg \text{Knows}(x, y))$
6. $\forall x \exists y \text{Friend}(x, y)$
7. $\forall x \forall y (\text{Person}(x) \wedge \text{Person}(y) \wedge \text{Criticise}(x, y) \rightarrow \neg \text{Friend}(x, y))$
8. $\text{Criticise}(\text{Alice}, \text{John})$
9. $\forall x \forall y \text{Criticise}(x, y) \rightarrow \text{Criticise}(y, x)$
10. Is this query true or false? $\neg \text{Friend}(\text{John}, \text{Alice})$

Symmetry Example

Recall example from previous lecture, now with a new sentence added:

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8. $\text{Criticise}(\text{Alice}, \text{John})$
9. $\forall x \forall y \text{Criticise}(x, y) \rightarrow \text{Criticise}(y, x)$
10. Is this query true or false? $\neg \text{Friend}(\text{John}, \text{Alice})$
 - Is John a Person?

Symmetry Example

Recall example from previous lecture, now with a new sentence added:

1. $\text{Prof}(\text{Alice})$
2. $\forall x (\text{Prof}(x) \rightarrow \text{Person}(x))$
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8. $\text{Criticise}(\text{Alice}, \text{John})$
9. $\forall x \forall y \text{Criticise}(x, y) \rightarrow \text{Criticise}(y, x)$
10. Is this query true or false? $\neg \text{Friend}(\text{John}, \text{Alice})$
 - John is a person; sentences 3, 4, and 2
 - Is Alice a person?

Symmetry Example

Recall example from previous lecture, now with a new sentence added:

1. $\text{Prof}(\text{Alice})$
2. $\forall x (\text{Prof}(x) \rightarrow \text{Person}(x))$
3. $\text{Dean}(\text{John})$
4. $\forall x (\text{Dean}(x) \rightarrow \text{Prof}(x))$
5. $\forall x \exists y (\text{Prof}(x) \wedge \text{Dean}(y) \rightarrow \text{Friend}(x, y) \vee \neg \text{Knows}(x, y))$
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8. $\text{Criticise}(\text{Alice}, \text{John})$
9. $\forall x \forall y \text{Criticise}(x, y) \rightarrow \text{Criticise}(y, x)$
10. Is this query true or false? $\neg \text{Friend}(\text{John}, \text{Alice})$
 - John is a person; sentences 3, 4, and 2
 - Alice is a person; sentences 1 and 2
 - Has John ever criticised Alice?

Symmetry Example

Recall example from previous lecture, now with a new sentence added:

1. $\text{Prof}(\text{Alice})$
2. $\forall x (\text{Prof}(x) \rightarrow \text{Person}(x))$
3. $\text{Dean}(\text{John})$
4. $\forall x (\text{Dean}(x) \rightarrow \text{Prof}(x))$
5. $\forall x \exists y (\text{Prof}(x) \wedge \text{Dean}(y) \rightarrow \text{Friend}(x, y) \vee \neg \text{Knows}(x, y))$
6. $\forall x \exists y \text{Friend}(x, y)$
7. $\forall x \forall y (\text{Person}(x) \wedge \text{Person}(y) \wedge \text{Criticise}(x, y) \rightarrow \neg \text{Friend}(x, y))$
8. $\text{Criticise}(\text{Alice}, \text{John})$
9. $\forall x \forall y \text{Criticise}(x, y) \rightarrow \text{Criticise}(y, x)$
10. Is this query true or false? $\neg \text{Friend}(\text{John}, \text{Alice})$
 - John is a person; sentences 3, 4, and 2
 - Alice is a person; sentences 1 and 2
 - John has criticised Alice; sentences 7, 8, and 9
 - Query result is true because sentence 7 is true

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Entailment

- ❑ *Validity, satisfiability, and unsatisfiability* are properties of individual sentences
- ❑ In logical reasoning, we are not so much concerned with individual sentences as we are with the **relationships** between sentences
- ❑ we would like to know, given some sentences (in KR the knowledge base), whether other sentences are or are not logical conclusions
- ❑ This relative property is known as *logical entailment*.

Entailment

- ❑ Formally, a set of sentences Δ logically entails a sentence ϕ (written $\Delta \models \phi$) if and only if every interpretation that satisfies Δ also satisfies ϕ .
 - For example, the set of sentences $\{p, p \wedge r\}$ logically entails the sentence $(p \vee q)$
 - Why?
- ❑ Inference: the process of calculating entailments
 - ❑ sound: get only entailments
 - ❑ complete: get all entailments
- ❑ Logic: study of entailment relations

Using Entailment Relations

- ☐ Find out if the truth of a sentence is implicit in what we already know
- ☐ Check if the sentence is **entailed** by the KB
- ☐ We need to determine whether every logical interpretation that satisfies KB also satisfies the sentence

Using Entailment Relations – Example 1

Example query: “Is there a company whose CEO loves Alice?”

$\exists x \text{ Company}(x) \wedge \text{Loves}(\text{ceoOf}(x), \text{Alice})$

Suppose $\zeta \models \text{KB}$

(Remember: Interpretation $\zeta = (D, I)$)

Then $\zeta \models \text{Rich}(\text{John}), \text{Man}(\text{John})$

and $\zeta \models \forall x \text{ Rich}(x) \wedge \text{Man}(x) \rightarrow \text{Loves}(x, \text{Alice})$

so $\zeta \models \text{Loves}(\text{John}, \text{Alice})$

Also $\zeta \models \text{John} = \text{ceoOf}(\text{IC})$

so $\zeta \models \text{Loves}(\text{ceoOf}(\text{IC}), \text{Alice})$

Finally $\zeta \models \text{Company}(\text{InsuranceCorporation})$

and $\zeta \models \text{IC} = \text{InsuranceCorporation}$

so $\zeta \models \text{Company}(\text{IC})$

Thus, $\zeta \models \text{Company}(\text{IC}) \wedge \text{Loves}(\text{ceoOf}(\text{IC}), \text{Alice})$

and so

$\zeta \models \exists x \text{ Company}(x) \wedge \text{Loves}(\text{ceoOf}(x), \text{Alice})$ is **true**

Activity 2 – Example 2

In pairs, solve the following problem (5 minutes) and then share your answers with the rest of your colleagues (5 minutes):

“If no man is blackmailing John, then is he being blackmailed by somebody he loves?”

$\forall x (Man(x) \rightarrow \neg Blackmails(x, John)) \rightarrow \exists y (Loves(John, y) \wedge Blackmails(y, John))$

Let $\zeta \models KB \cup \{\forall x (Man(x) \rightarrow \neg Blackmails(x, John))\}$

Show $\zeta \models \exists y (Loves(John, y) \wedge Blackmails(y, John))$

Summary

- Introduced knowledge engineering through a soap opera example
- Discussed the importance of vocabulary, facts, and entailment relations in knowledge engineering

Until the next session:

- Install SWI-Prolog to your computer.